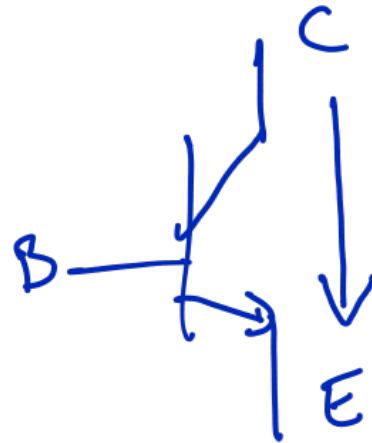
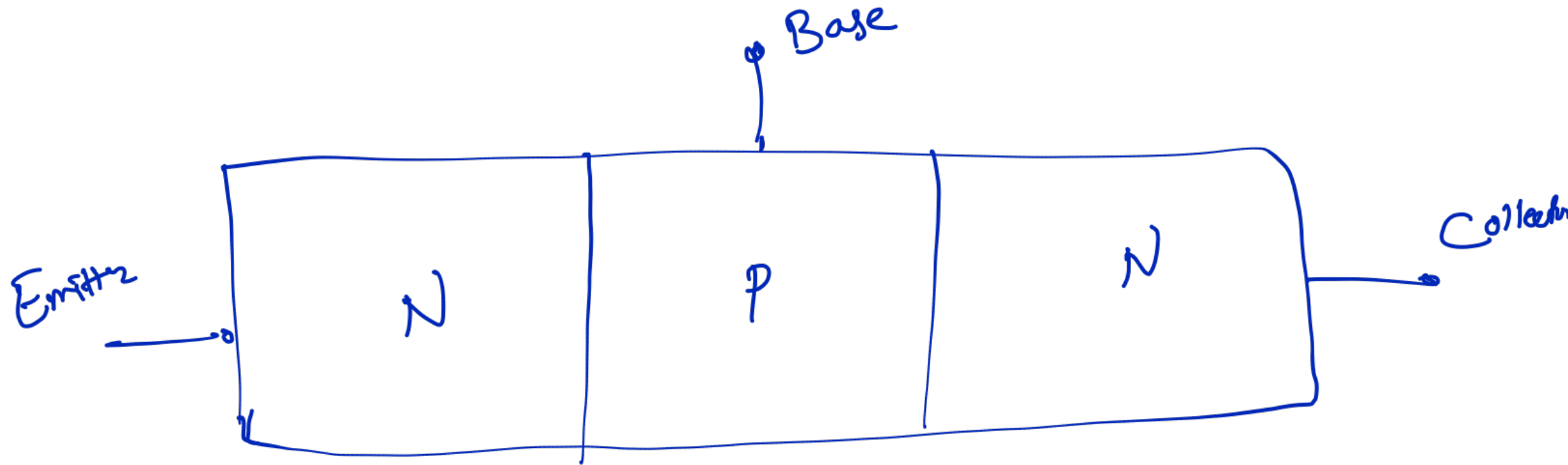
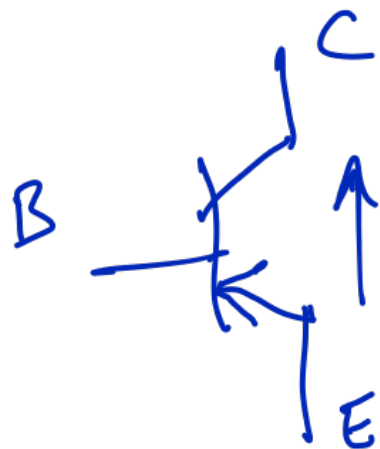
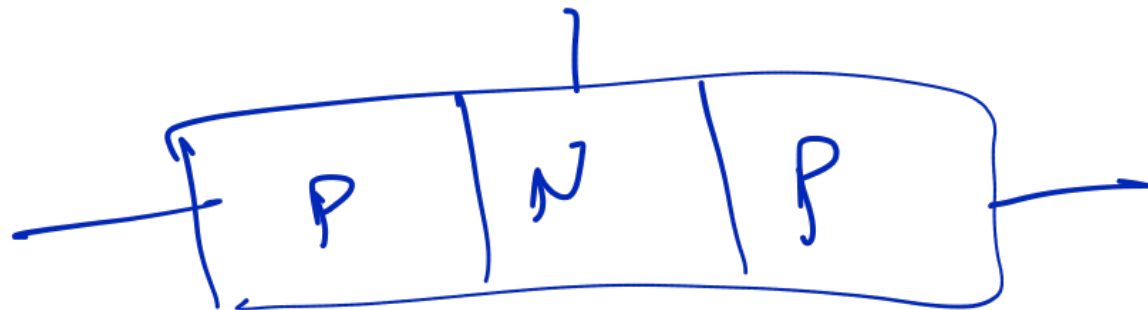
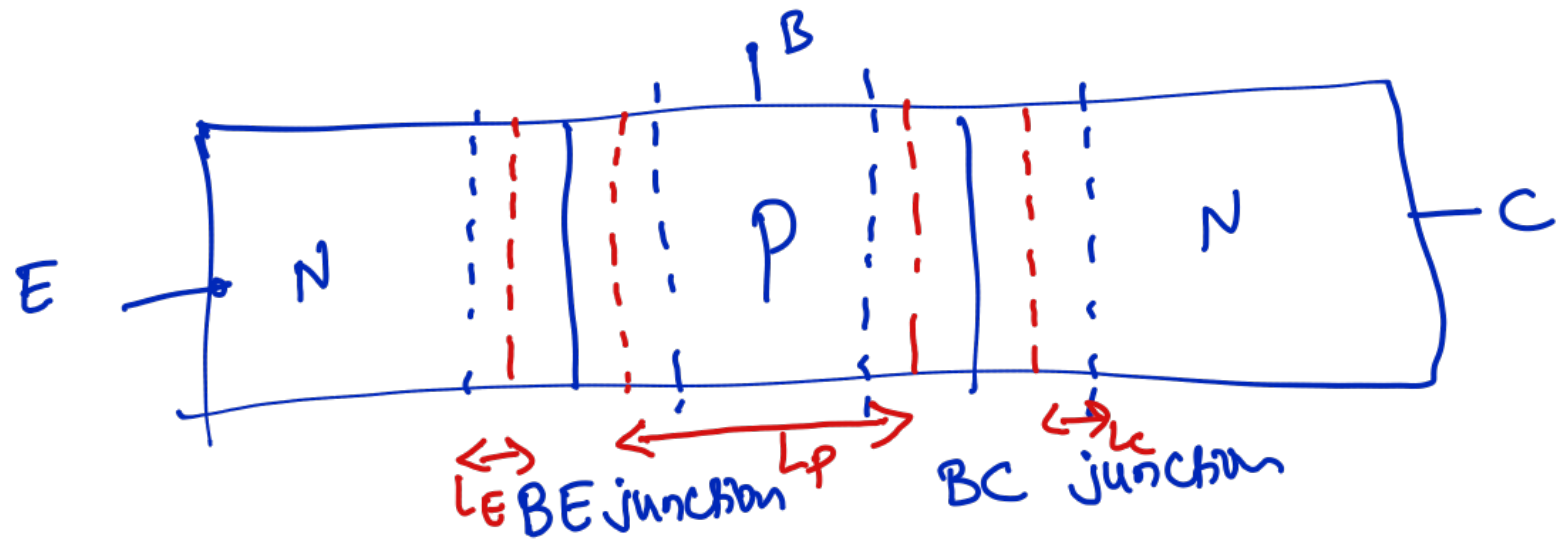


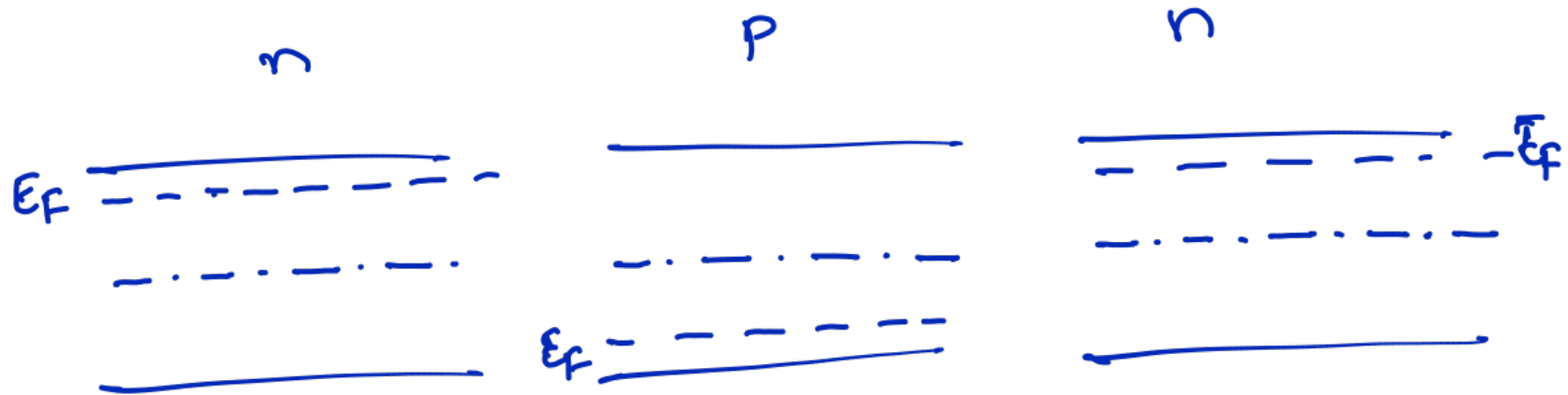
→ Reliability??

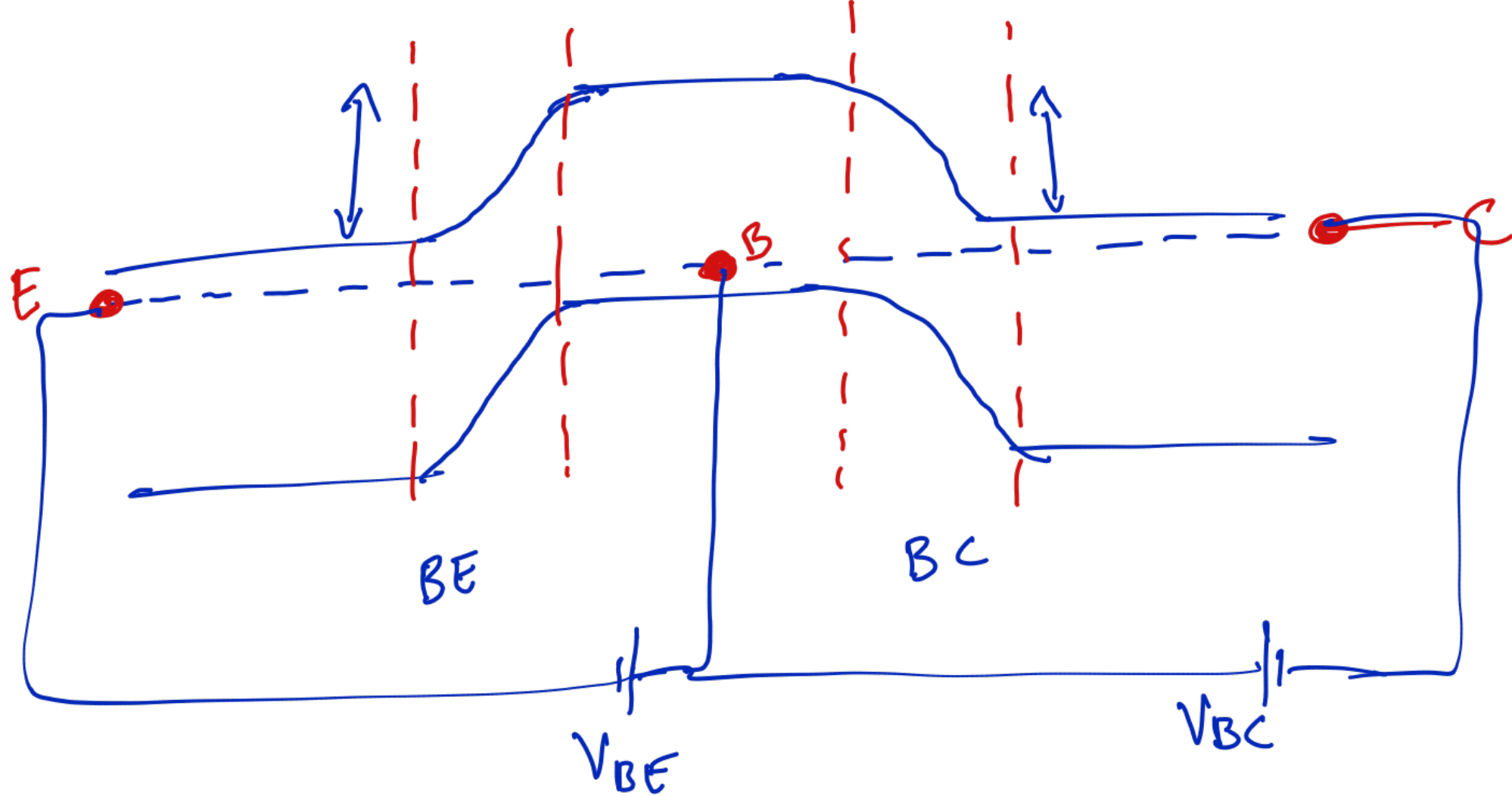




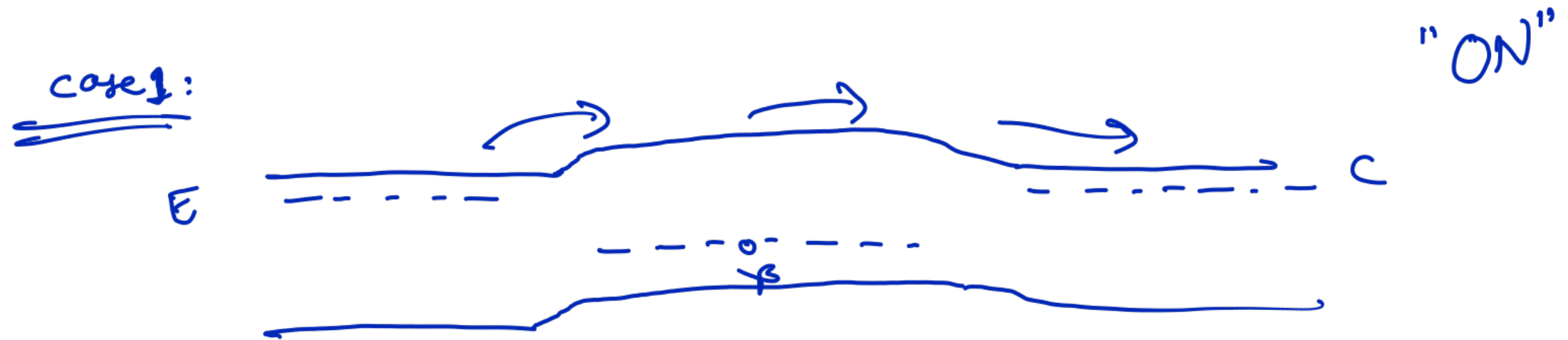


- Q1 what should be length of L_p ?
- Q2 what should be the dopings (N_A, N_D) ?

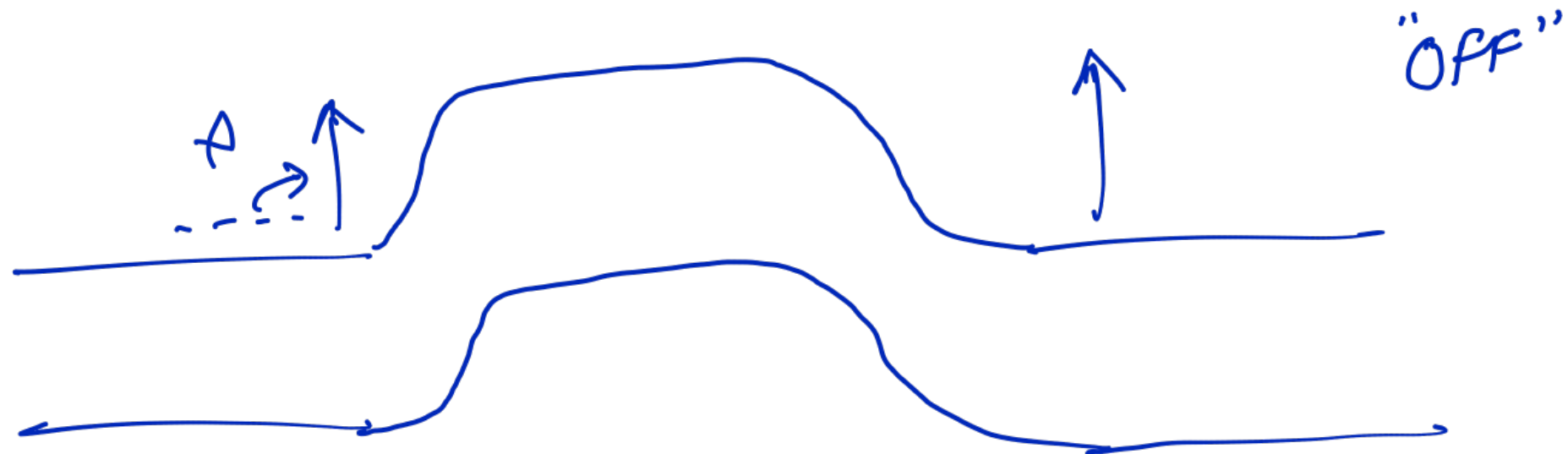




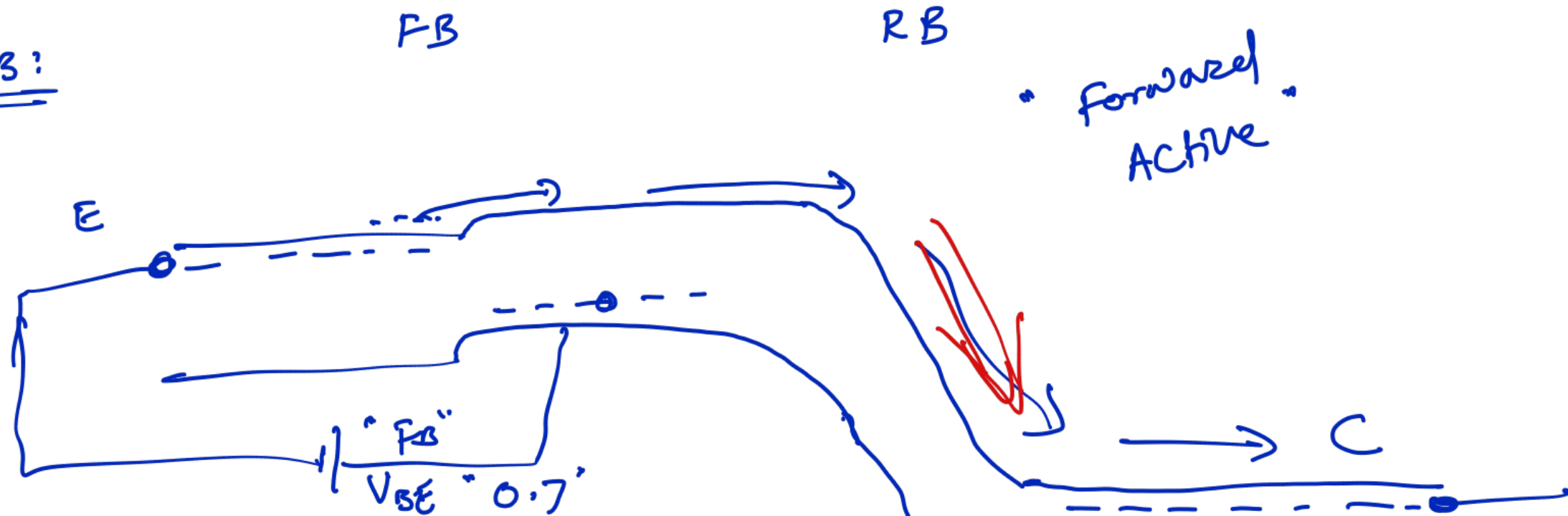
	BE	BC	<u>Saturation</u>	
case 1:	FB	FB	"ON"	"Switch Applications"
			"OFF"	
case 2:	RB	RB	Active	} Amplifier
case 3:	FB	RB	Reverse Active	
case 4:	RB	FB		} Not used

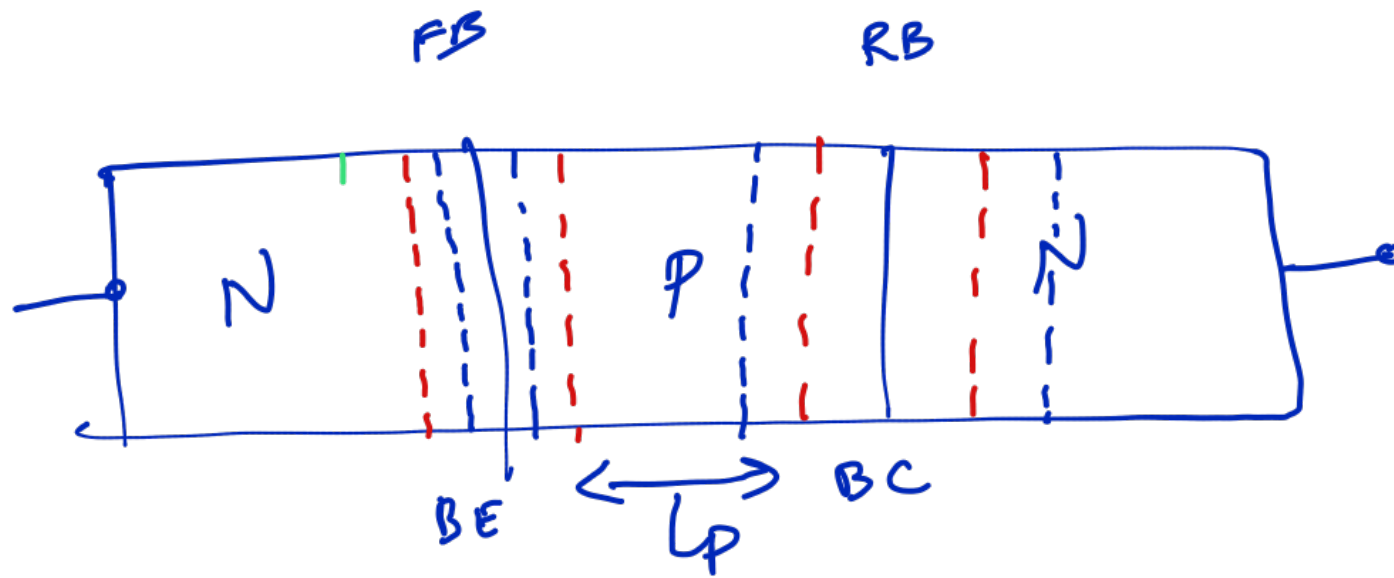


Case 2:

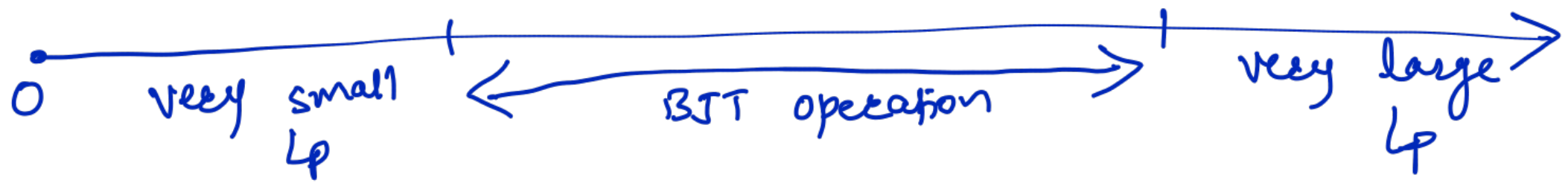


Case 3:





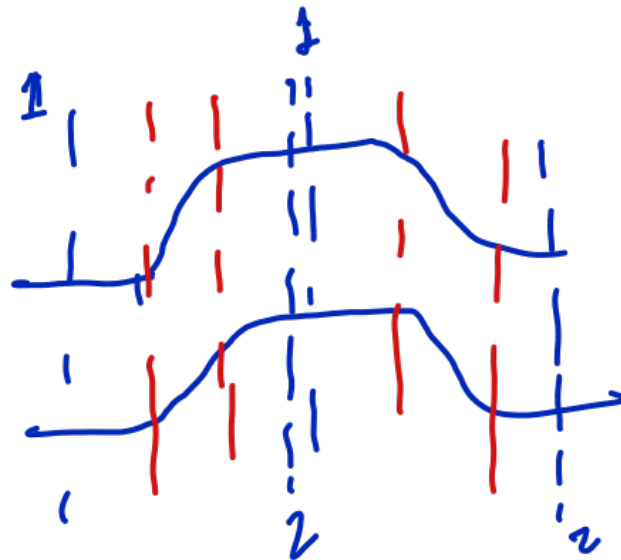
L_p



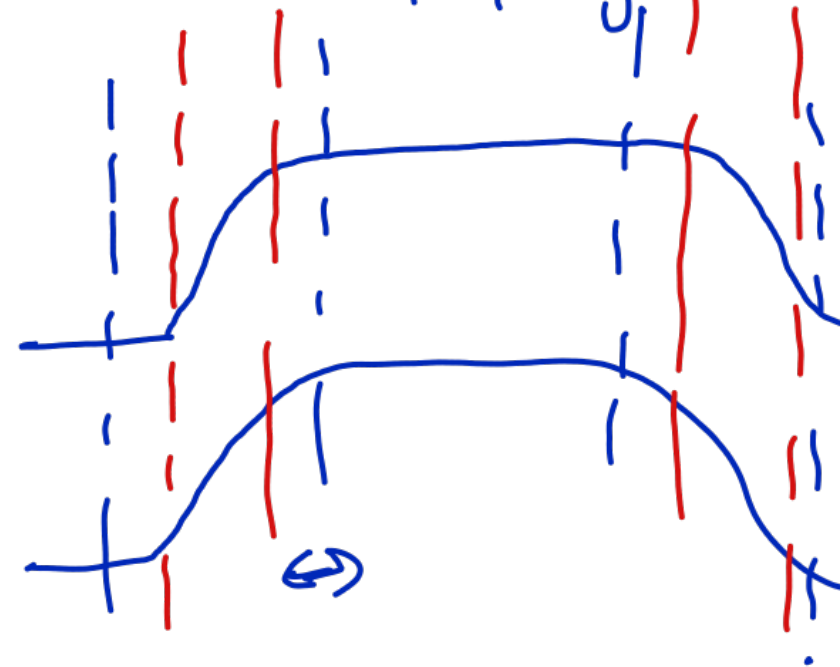
- Depletion Regions are merging

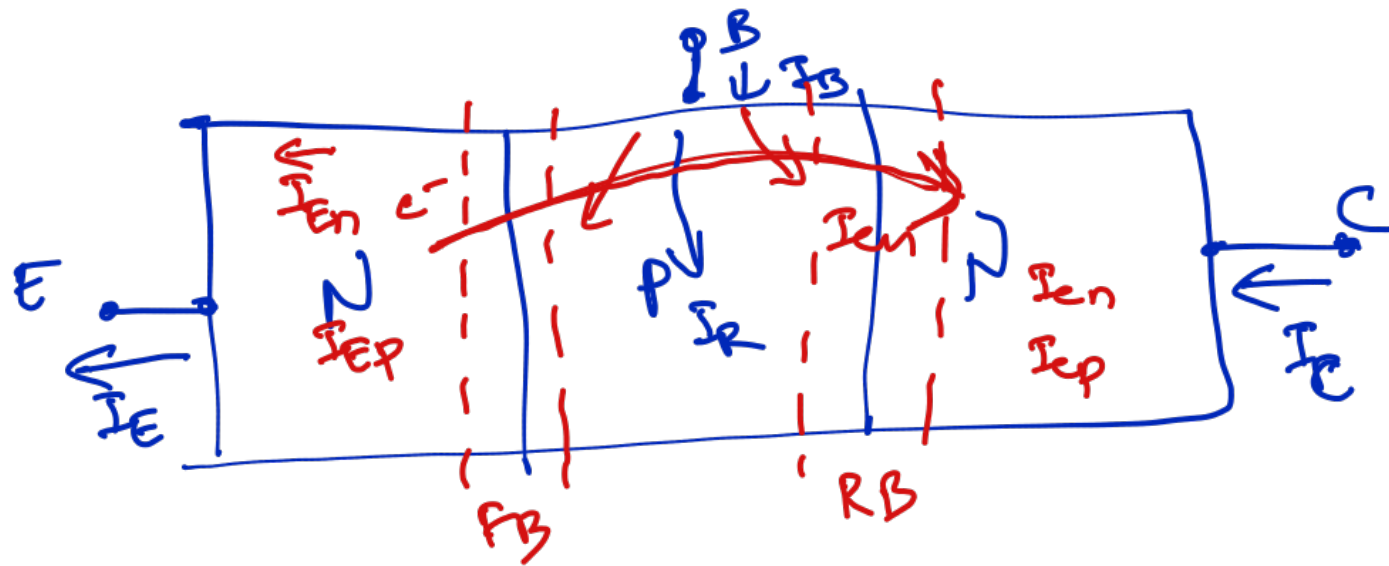


- Diffusion regions merge inside p-region



- Behaves as separate diodes
- Separate diffusion lengths in p-region





$$P_E = I_B + I_C$$

$$I_E = I_{En} + I_{EP}$$

$$I_c = I_{cn} + \underline{I_{cp}} \rightarrow 0$$

$$I_B = I_{EP} + \underbrace{I_{EP}}_{\rightarrow 0} + \underbrace{I_R}_{\rightarrow 0}$$

$$I_c(O/P) \cong I_{cn}$$

$$I_B(i|p) \approx I_{ep}$$

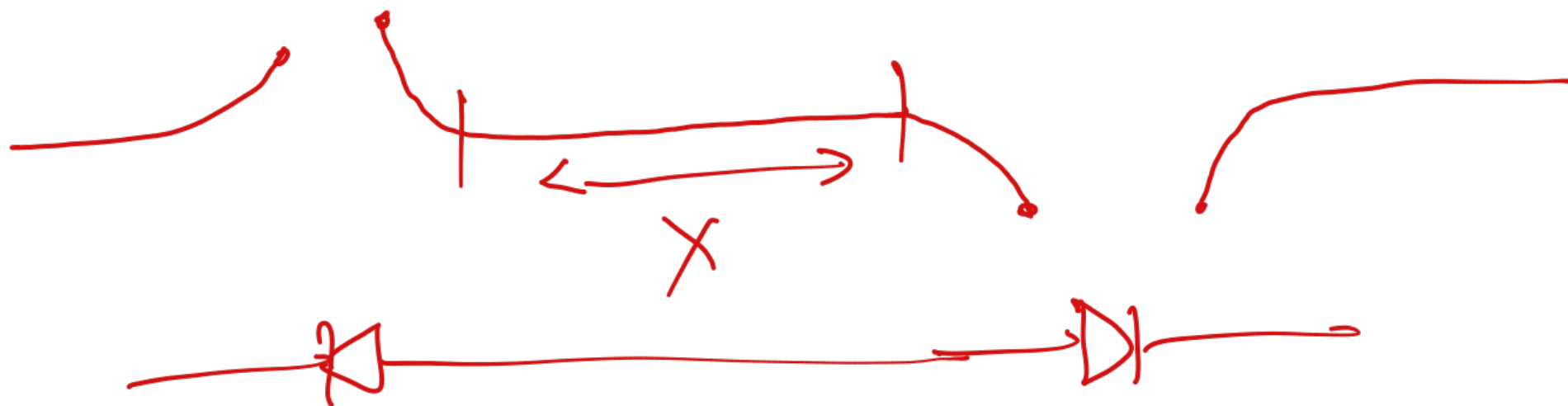
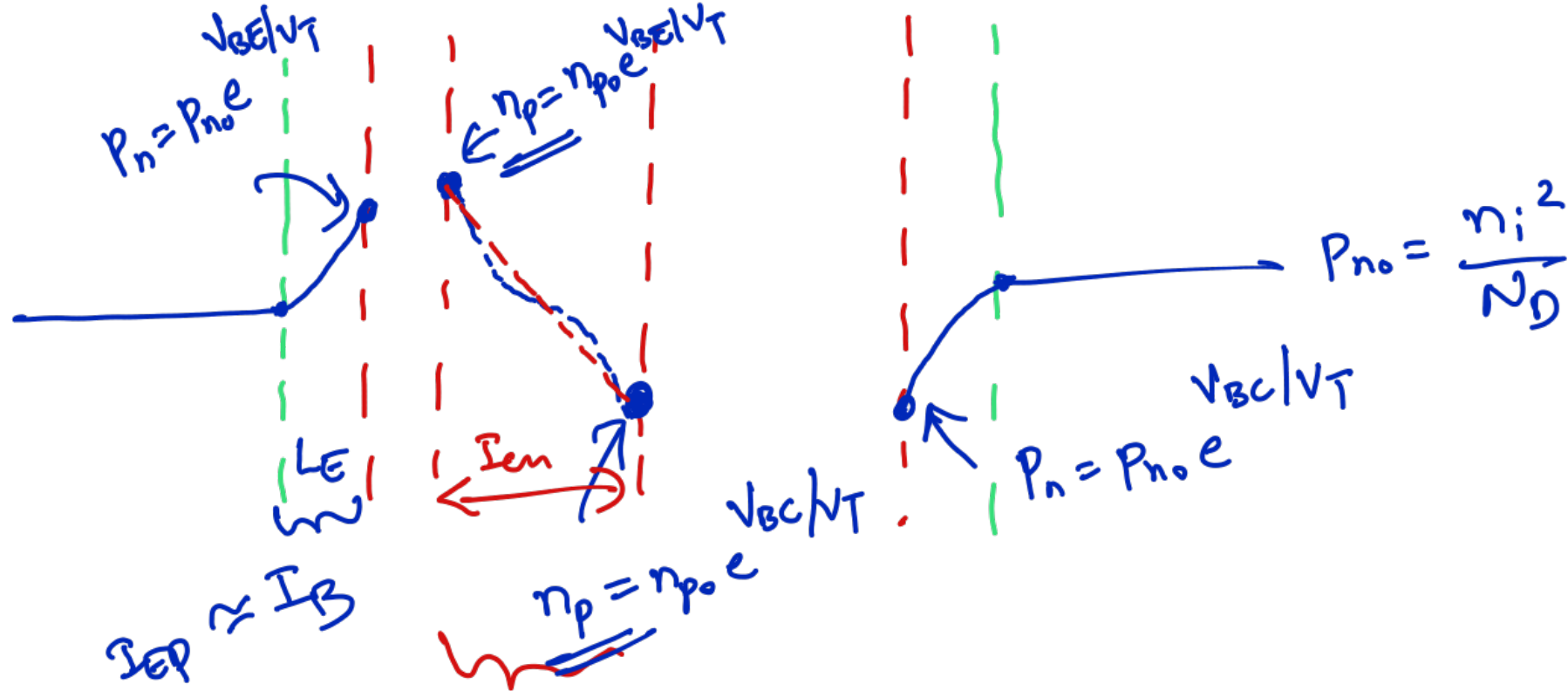
$$\frac{BE}{FB}$$
$$\frac{BC}{RB}$$

2

9

2

$$\frac{n_i^2}{N_D} = P_{n0}$$



$$I_B \approx I_{EP}$$

$$I_{EP} \approx -q D_p \frac{dp}{dx}$$

$$= q D_n \frac{p_{n0} e^{V_{BE}/V_T} - p_{n0}}{L_E}$$

$$I_B = \frac{q D_n p_{n0}}{L_E} \left(e^{V_{BE}/V_T} - 1 \right)$$

↓
0

"FB"

$$I_C \approx I_{en}$$

$$I_{en} = q D_n \frac{dn}{dx}$$

$$= q D_n \frac{n_{p0} e^{V_{BE}/V_T} - n_{p0} e^{V_{BC}/V_T}}{L_p}$$

$$I_C = \frac{q D_n}{L_p} n_{p0} \left(e^{V_{BE}/V_T} - e^{V_{BC}/V_T} \right)$$

↘
0

BE

FB

BC

RD

$$I_B = \frac{q D_p P_{n0}}{L_E} (e^{V_{BE}/V_T})$$

(i/p)

$$I_C = \frac{q D_n}{L_p} n_{p0} (e^{V_{BE}/V_T})$$

(o/p)

Current Gain = $\frac{I_C}{I_B} = \frac{D_n n_{p0} \cdot L_E}{L_p D_p P_{n0}} = \frac{\overbrace{D_n \cdot L_E \cdot N_D}^{\sim \text{fixed}}}{\underbrace{D_p \cdot L_p \cdot N_A}_{\uparrow n_i^2/N_D}}$

" β "

" β varies in the range 10, 100, 1000..."

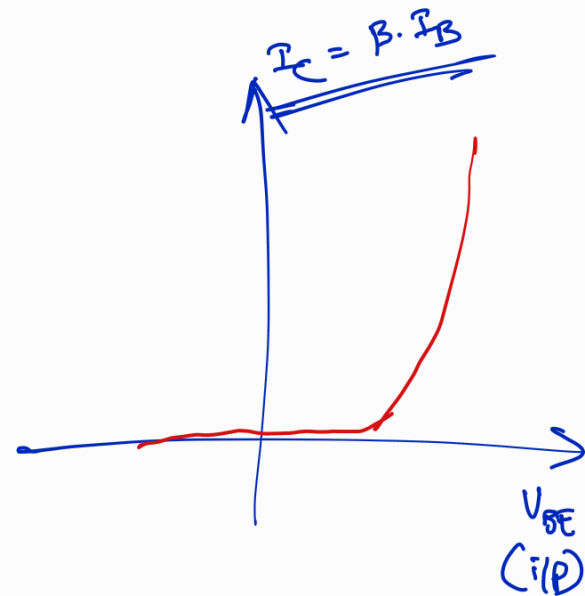
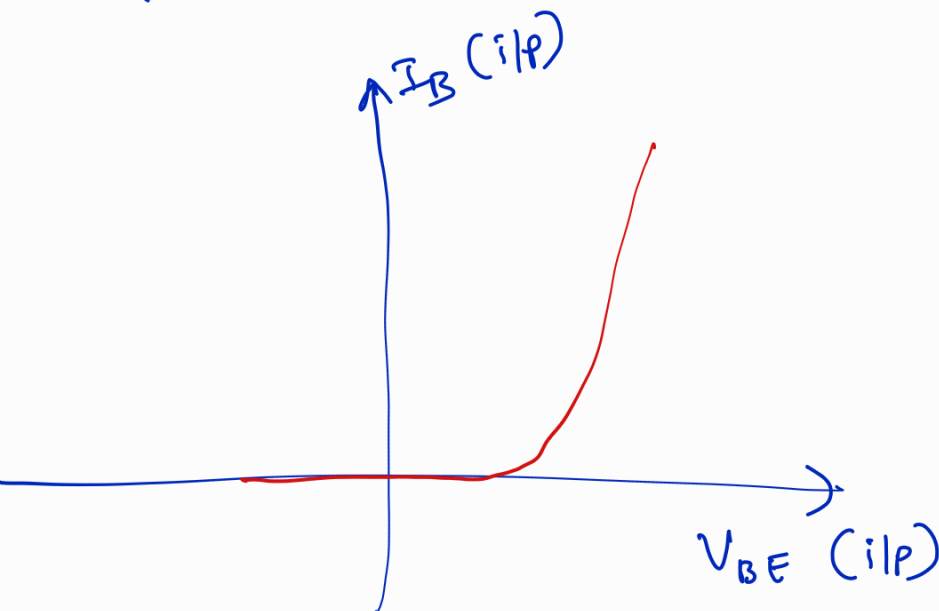
$\beta \uparrow$

$\Rightarrow$

$L_p \downarrow$,
limitation

$N_D \gg N_A$

"CE" highly than Base Region.



Bipolar Junction Transistor (BJT)

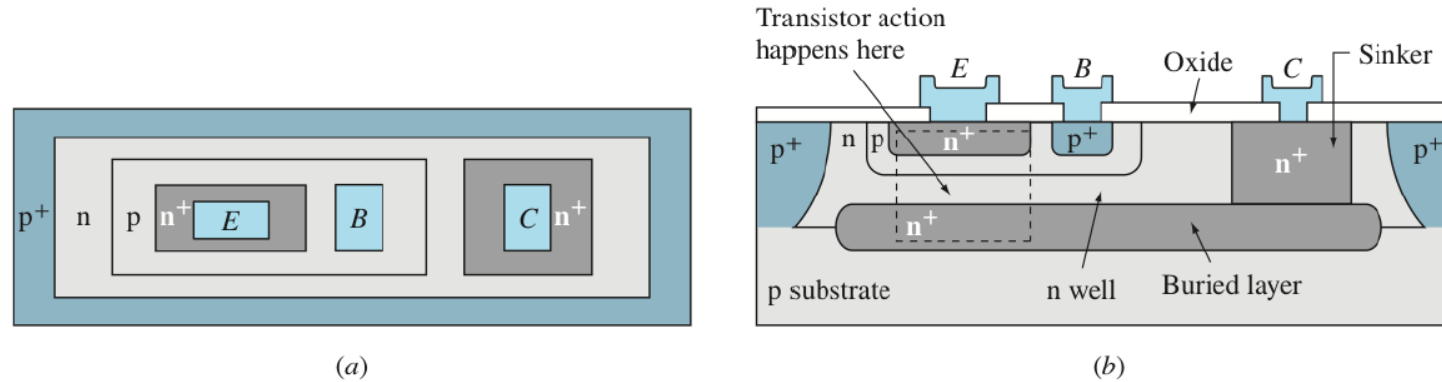
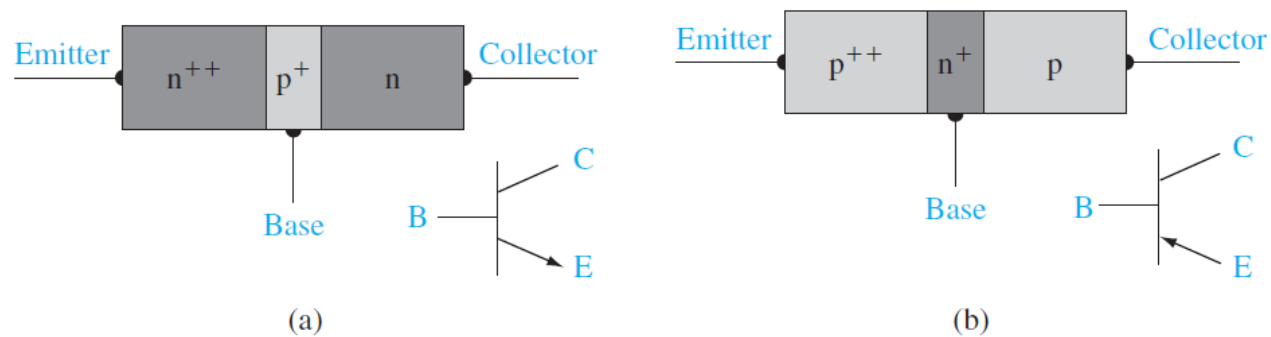
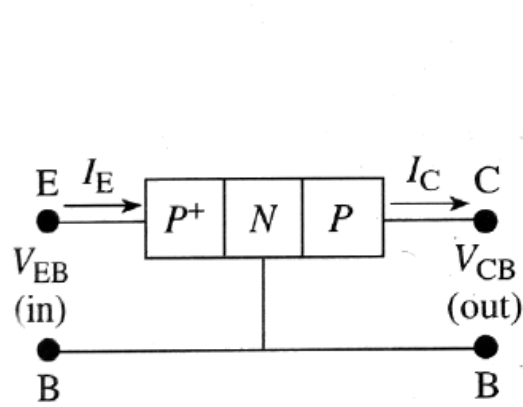


Figure IV.1 Schematic diagram of an npn bipolar transistor used in integrated circuits. (a) Top view and (b) cross-sectional view.



FET	BJT
Source	Emitter
Drain	Collector
Gate	Base

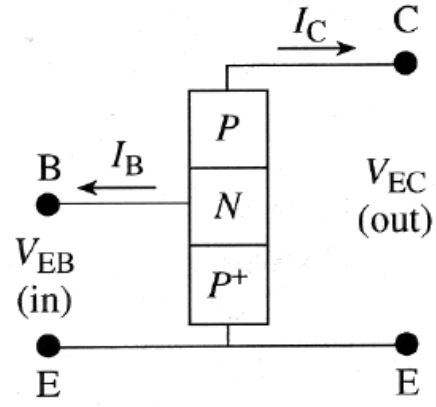
BJT - Configurations



Common base

Both the input and output share the base “in common”.

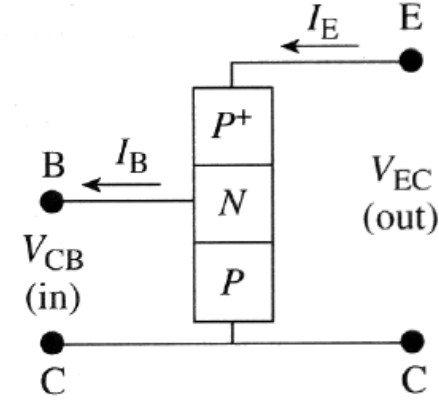
Rarely used. Exceptions include Cascode circuits and voltage protection.



Common emitter

Both the input and output share the emitter “in common”.

Widely Used. For example gain stages in amplifiers.

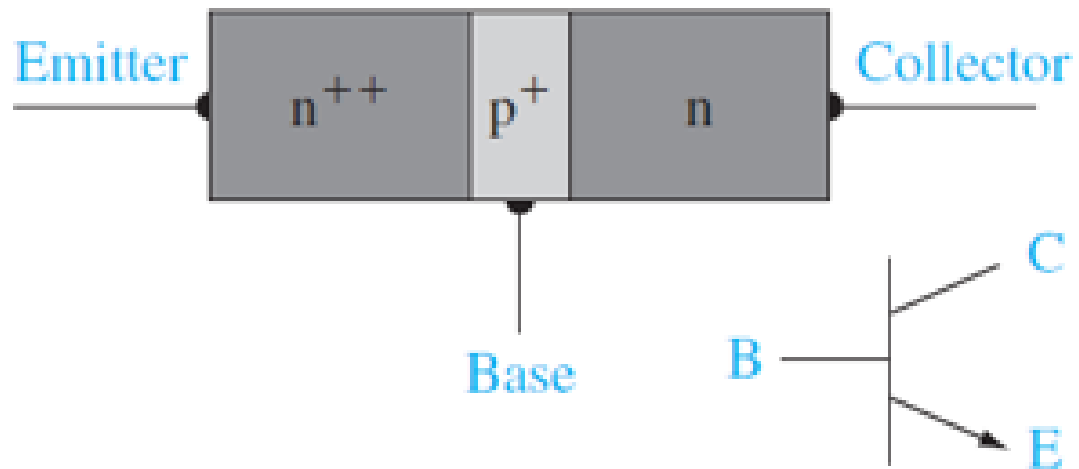


Common collector

Both the input and output share the Collector “in common”.

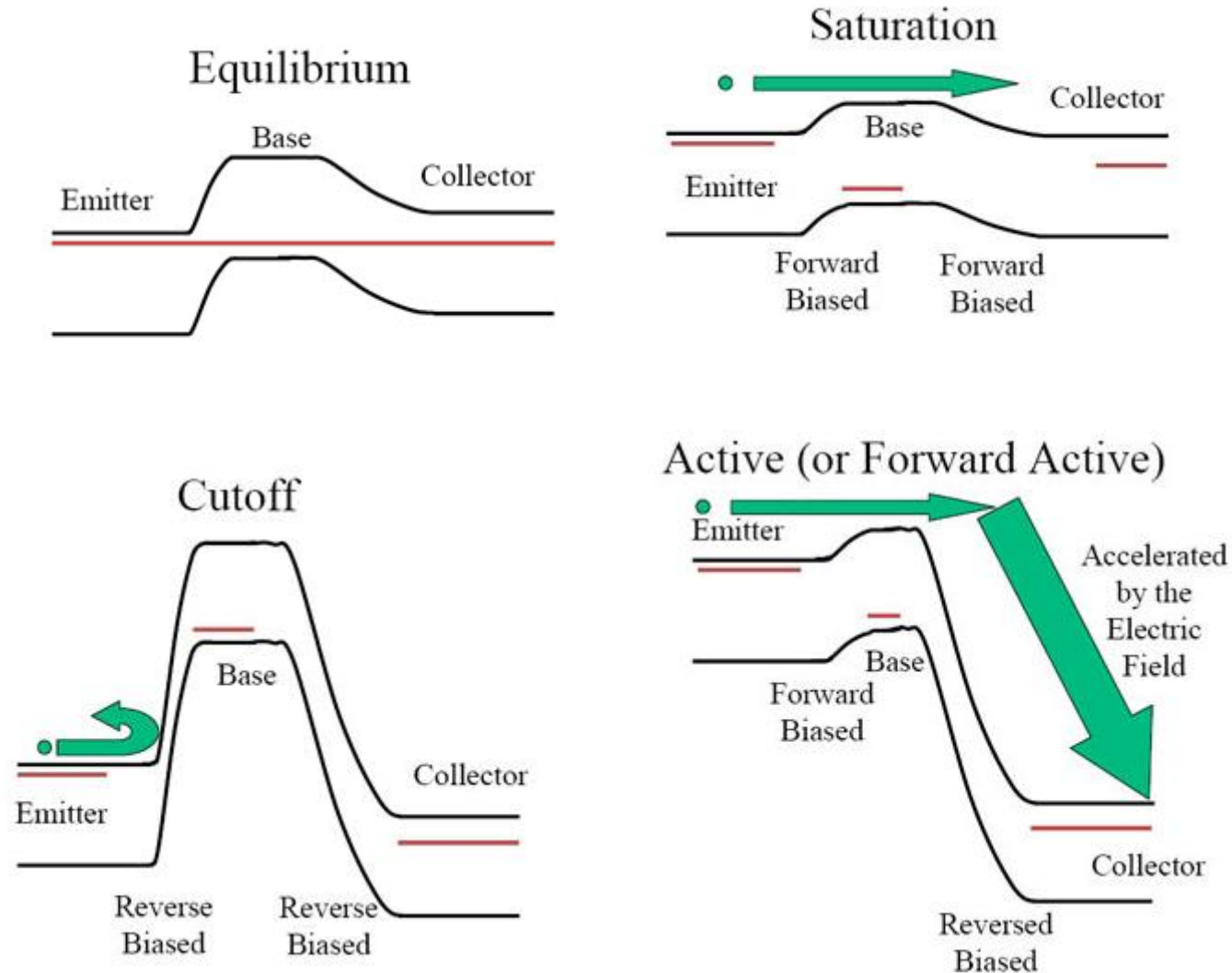
Widely Used. For example impedance transformation or buffer stages in amplifiers.

BJT – Modes of operation



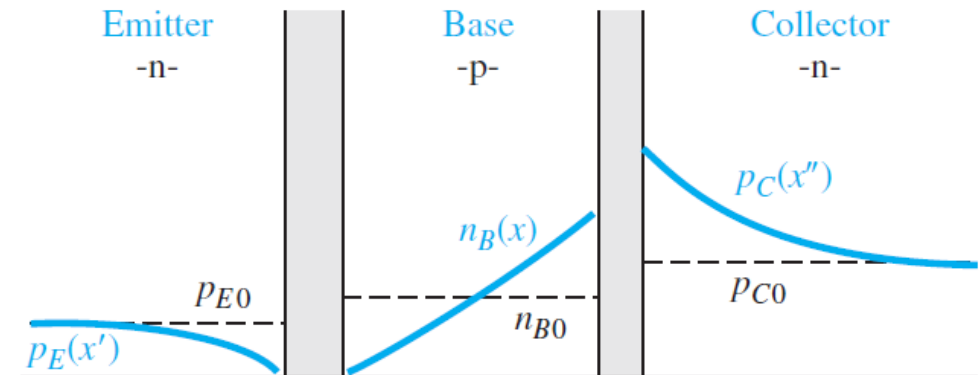
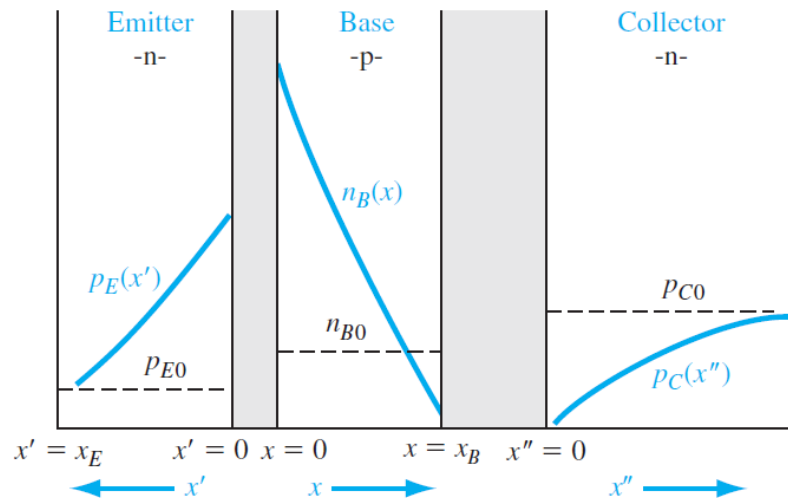
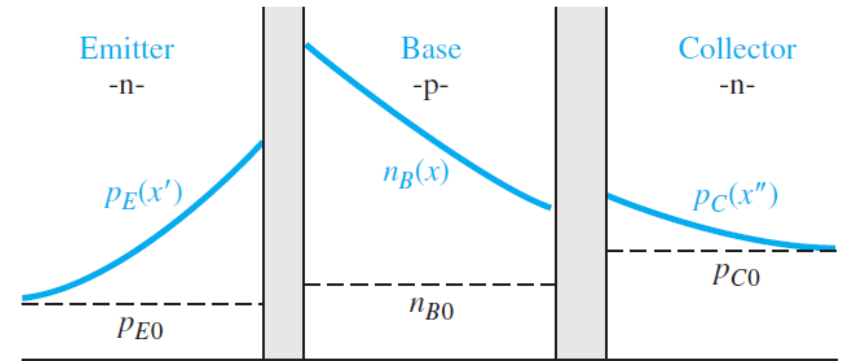
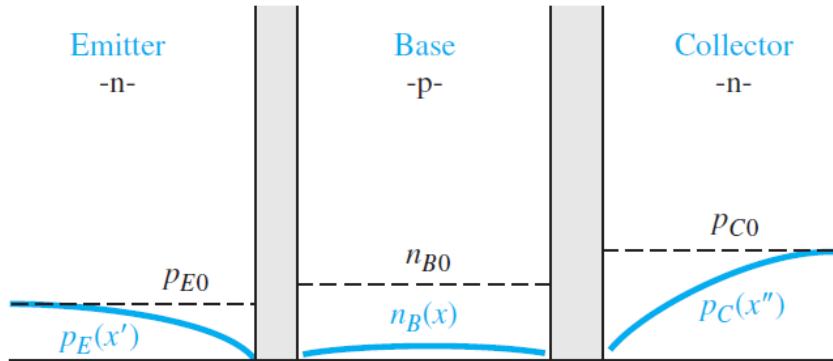
Region of operation	Base Emitter (B-E) junction	Base Collector (B-C) junction	Application
Cut-off	Reverse biased	Reverse biased	BJT as switch
Forward active	Forward biased	Reverse biased	BJT as amplifier
Inverse active	Reverse biased	Forward biased	Not much used
Saturation	Forward biased	Forward biased	BJT as switch

BJT – Different Modes of operation, Band Diagrams

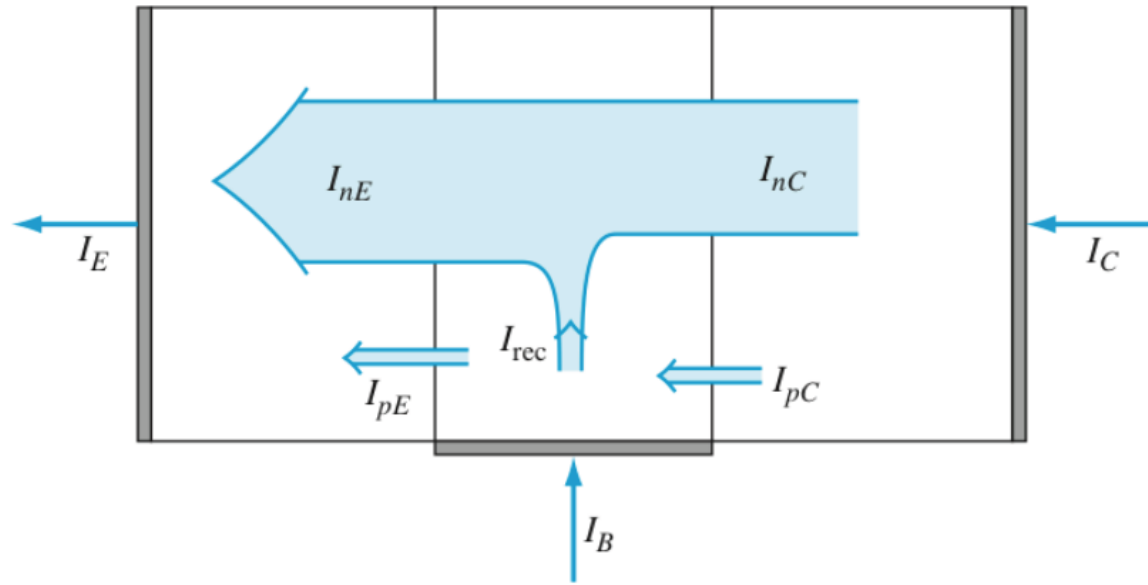


Minority carrier distribution for different modes of operation

Guess the biasing modes?



BJT – Currents – Forward Active



Collector Current

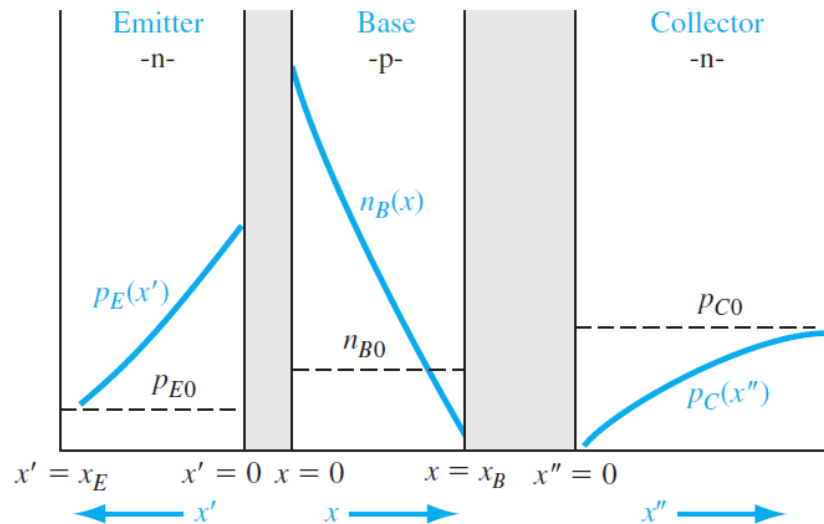
$$I_C = I_{nC} + I_{pC}$$

Base Current

$$I_B = I_{pE} + I_{pC} + I_{rec}$$

Emitter Current

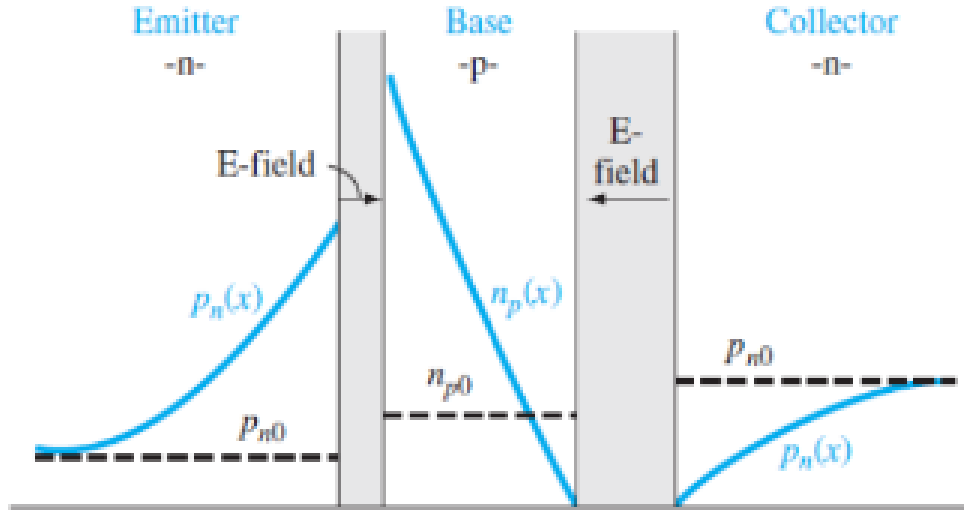
$$I_E = I_{nE} + I_{pE}$$



BJT – Forward Active - Currents

Collector Current

$$J_{n,C} = qD_n \frac{dn_p}{dx}$$



$$I_{n,C} = qAD_n \frac{n_p(0) - n_p(W_B)}{0 - W_B}$$

$$= -\frac{qAD_n n_{p0}}{W_B} \left[\exp\left(\frac{V_{BE}}{V_T}\right) - \exp\left(\frac{V_{BC}}{V_T}\right) \right]$$

Base Current

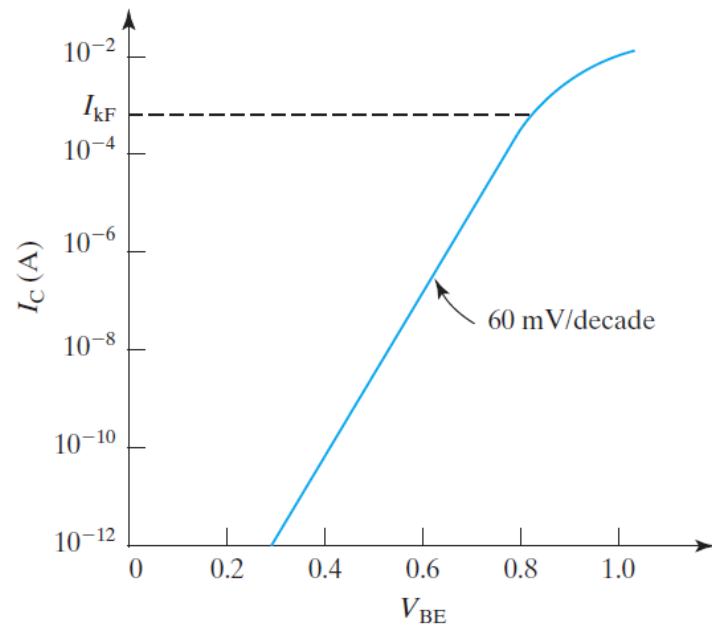
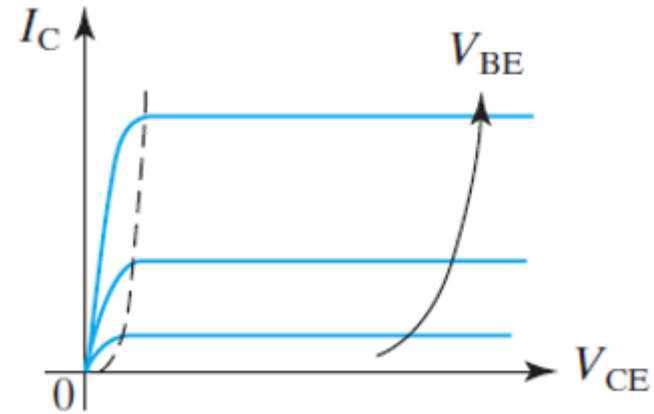
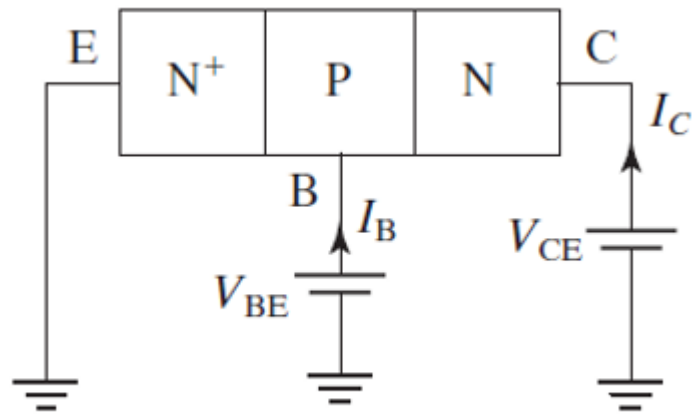
$$J_{p,E} = qD_p \frac{dp_n}{dx}$$

$$\begin{aligned} I_C &\approx -\frac{qAD_n n_{p0}}{W_B} \left[\exp\left(\frac{V_{BE}}{V_T}\right) \right] \approx -\frac{qAD_n n_{i,B}^2}{W_B N_B} \left[\exp\left(\frac{V_{BE}}{V_T}\right) \right] \\ I_B &\approx -\frac{qAD_p p_{n0}}{W_E} \left[\exp\left(\frac{V_{BE}}{V_T}\right) \right] \approx -\frac{qAD_p n_{i,E}^2}{W_E N_E} \left[\exp\left(\frac{V_{BE}}{V_T}\right) \right] \end{aligned}$$

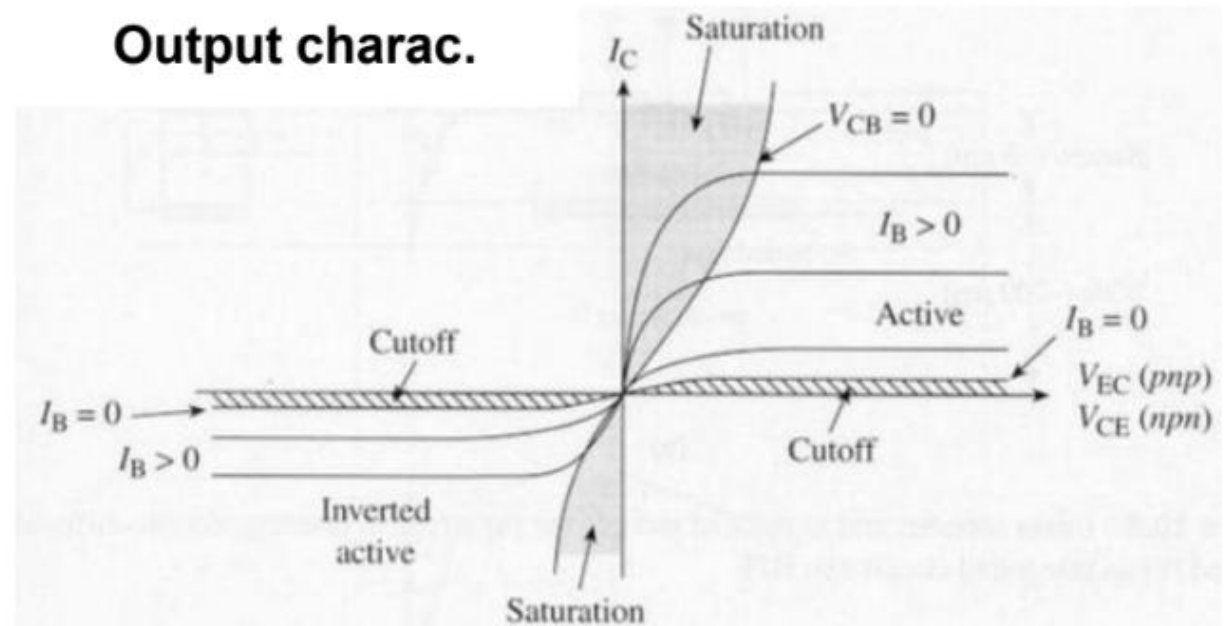
$$I_{p,E} = qAD_p \frac{p_n(0) - p_n(W_E)}{0 - W_E}$$

$$= -\frac{qAD_p p_{n0}}{W_E} \left[\exp\left(\frac{V_{BE}}{V_T}\right) - 1 \right]$$

BJT – CE Input and output characteristics



Output charac.

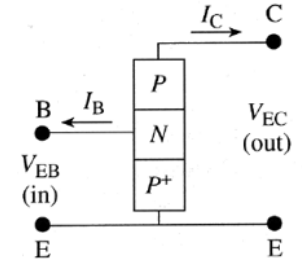


BJT – Performance Parameters

Current Gain (CE Configuration): Amount of current at output (C) for the given the amount of current injected (B)

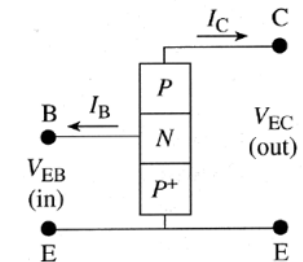
$$\beta = \frac{I_{out}}{I_{in}} = \frac{I_C}{I_B}$$

$$\begin{aligned} I_C &\approx -\frac{qAD_n n_{p0}}{W_B} \left[\exp\left(\frac{V_{BE}}{V_T}\right) \right] \approx -\frac{qAD_n n_{i,B}^2}{W_B N_B} \left[\exp\left(\frac{V_{BE}}{V_T}\right) \right] \\ I_B &\approx -\frac{qAD_p p_{n0}}{W_E} \left[\exp\left(\frac{V_{BE}}{V_T}\right) \right] \approx -\frac{qAD_p n_{i,E}^2}{W_E N_E} \left[\exp\left(\frac{V_{BE}}{V_T}\right) \right] \\ \beta &= \frac{I_C}{I_B} = \frac{D_n n_{i,B}^2 W_E N_E}{D_p n_{i,E}^2 W_B N_B} \end{aligned}$$



Emitter Efficiency: Characterizes how effective the large electron current is controlled by the small hole current. Unity is best, zero is worst.

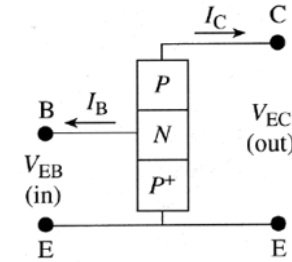
$$\gamma = \frac{I_{E_n}}{I_E} = \frac{I_{E_n}}{I_{E_n} + I_{E_p}}$$



BJT – Performance Parameters

Emitter Transport/injection Efficiency: Characterizes how much of the injected electron current is lost to recombination in the base. Unity is best, zero is worst.

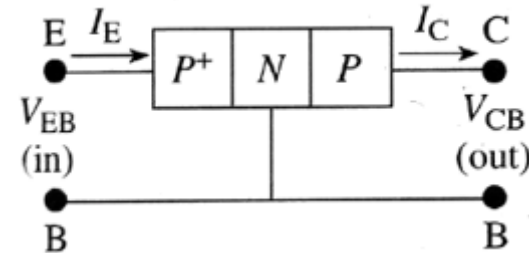
$$\alpha_T = \frac{I_{nC}}{I_{nE}} = \frac{I_{nE} - I_{rec}}{I_{nE}}$$



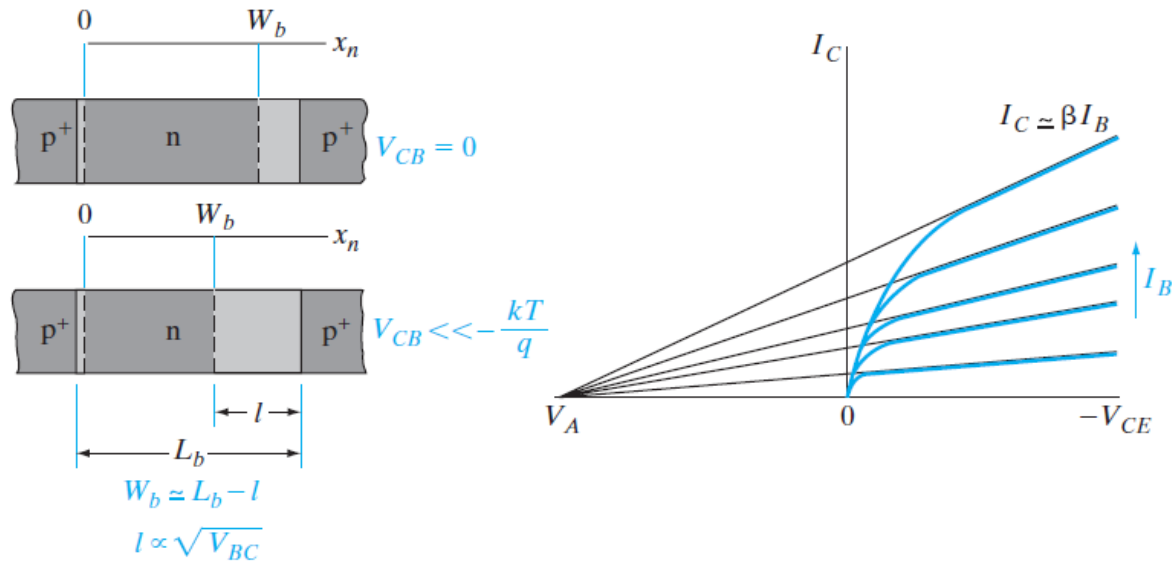
Current Gain (CB): Amount of current at output (C) for the given the amount of current injected (E)

$$\alpha = \frac{I_C}{I_E}$$

$$I_E = I_C + I_B$$
$$\alpha = \frac{\beta}{\beta + 1}$$



BJT – Non-idealities – 1. Base narrowing/modulation – Early Effect



$$I_C \approx -\frac{qAD_n n_{p0}}{W_B} \left[\exp\left(\frac{V_{BE}}{V_T}\right) \right] \approx -\frac{qAD_n n_{i,B}^2}{W_B N_B} \left[\exp\left(\frac{V_{BE}}{V_T}\right) \right]$$

$$I_B \approx -\frac{qAD_p p_{n0}}{W_E} \left[\exp\left(\frac{V_{BE}}{V_T}\right) \right] \approx -\frac{qAD_p n_{i,E}^2}{W_E N_E} \left[\exp\left(\frac{V_{BE}}{V_T}\right) \right]$$

$$\beta = \frac{I_C}{I_B} = \frac{D_n n_{i,B}^2 W_E N_E}{D_p n_{i,E}^2 W_B N_B}$$

BJT – Non-idealities – 2. Avalanche Breakdown

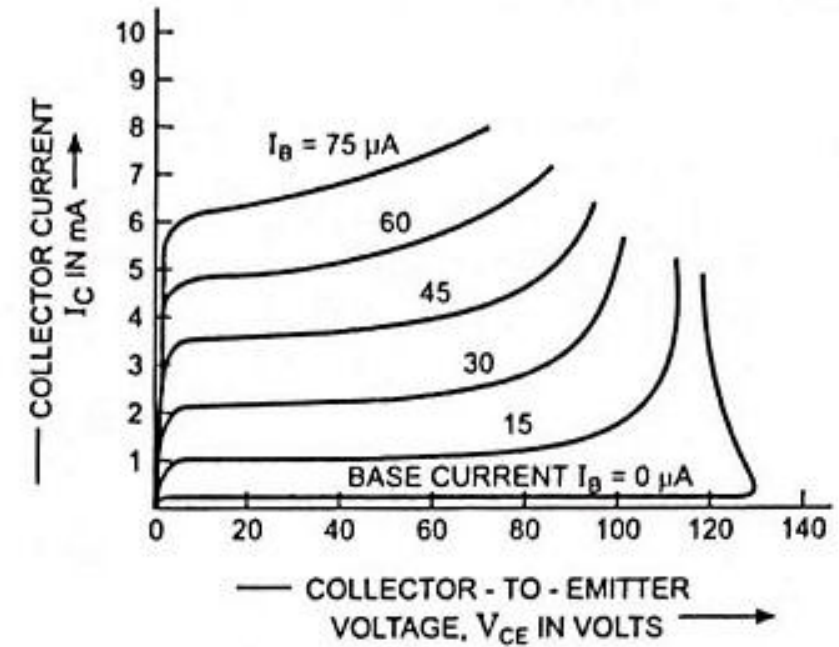
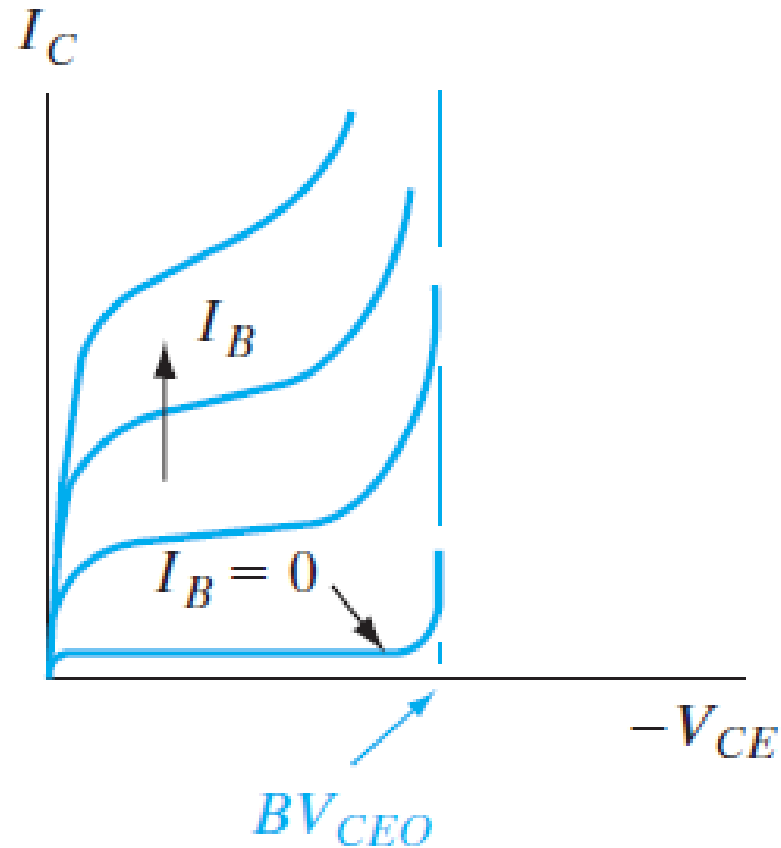
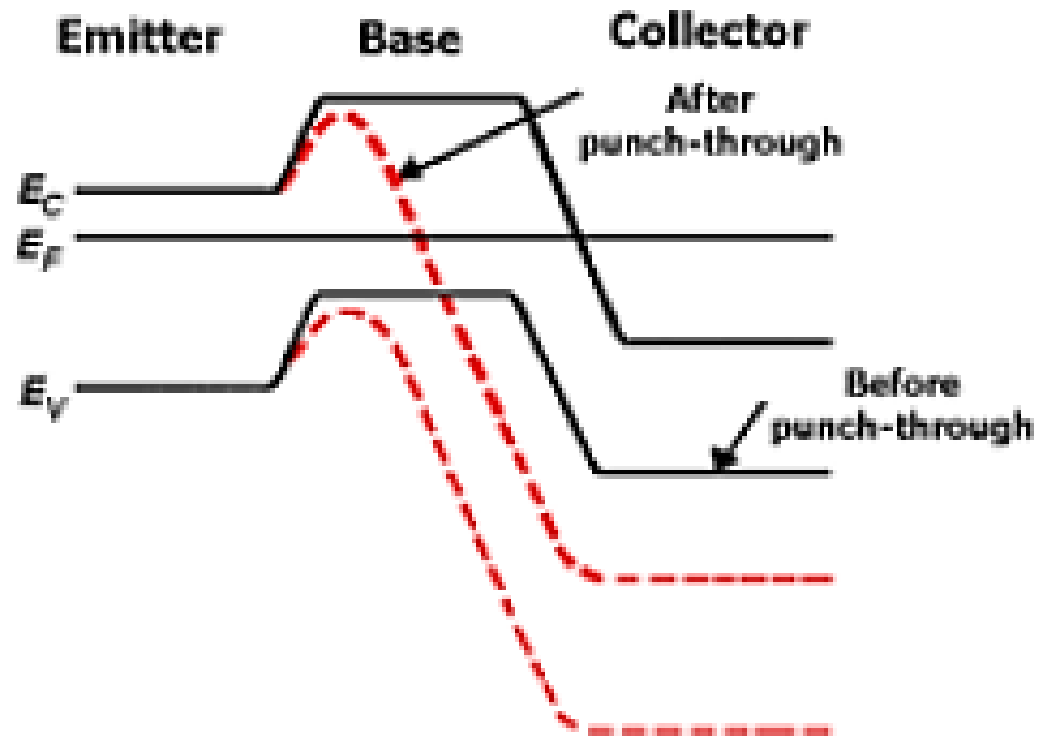


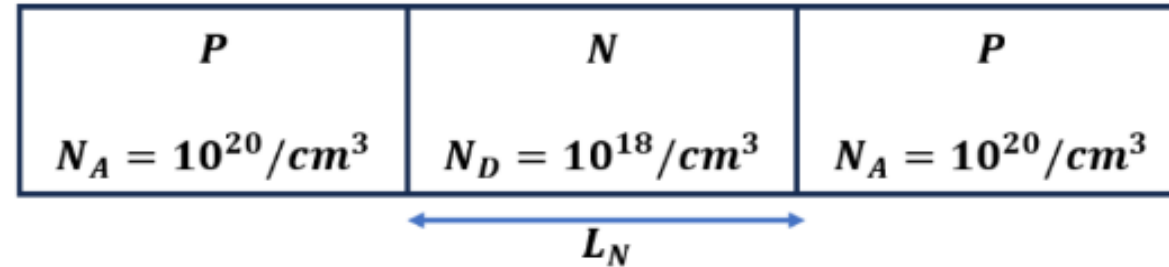
Fig. 26.76 Typical Collector Current Versus Collector-Emitter Voltage Characteristics of a BJT Showing Breakdown Effects

BJT – Non-idealities – 3. Punch Through Effect



BJT – Non-idealities – 3. Punch Through Effect

For a PNP structure shown below, assume the N-region length (L_N) is sufficiently large so that the depletion regions do not merge from two junctions. Assume Depletion approximation.

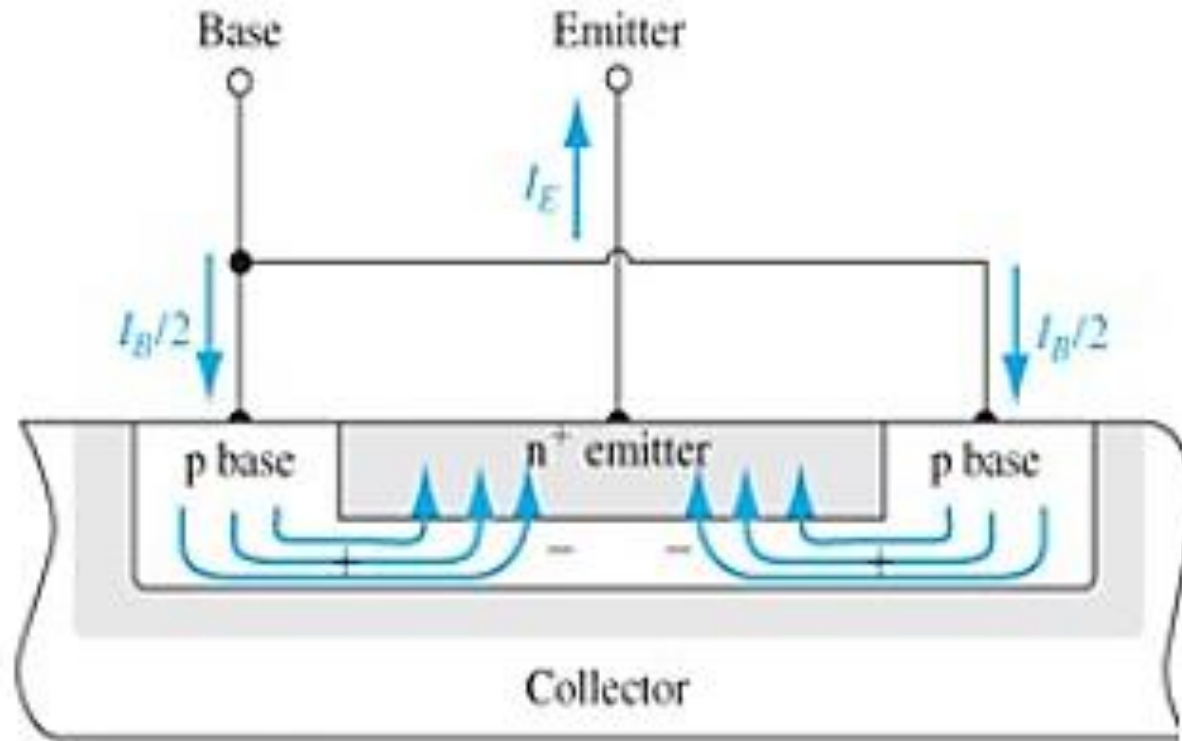


- a. Calculate the potential barrier value (1)
- b. Calculate depletion width (in nm) in P-region and N-region. Which region has larger portion of total depletion width? (2)
- c. Draw Band diagram, potential, E-field and charge profile. (4)

Now, assume $L_N = 30nm$. You will observe that the depletion regions from both sides merge inside N-region.

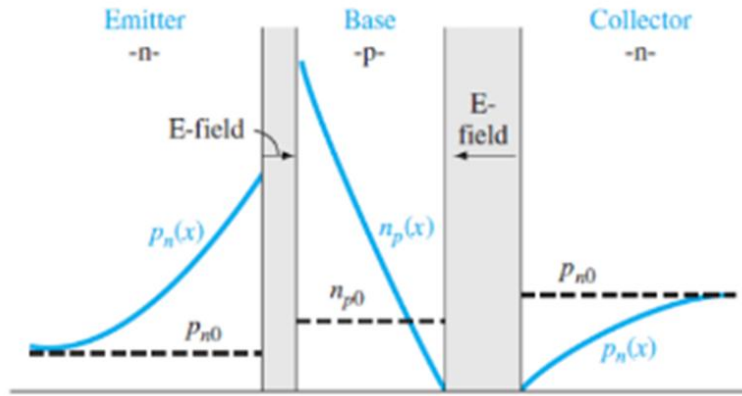
- d. Derive the equation for the new potential barrier and show its dependence on N_D, L_N (3)
- e. Calculate the new potential barrier value. (2)
- f. Draw the Band diagram, potential, E-field and charge profile when $L_N = 30nm$. Mark peak barrier value, peak electric field value. (4)
- g. With $L_N = 30nm$, to achieve the peak potential barrier of 0.5V, what should be N_D ? (2)
- h. With $N_D = 10^{18}/cm^3$, to achieve the peak potential barrier of 0.5V, what should be the L_N in nm? (2)

BJT – Non-idealities – 3. Current Crowding



More: High level injection/Thermal Effect, Bandgap narrowing, current crowding etc

Need for Heterojunction Bipolar Transistor (HBT)?



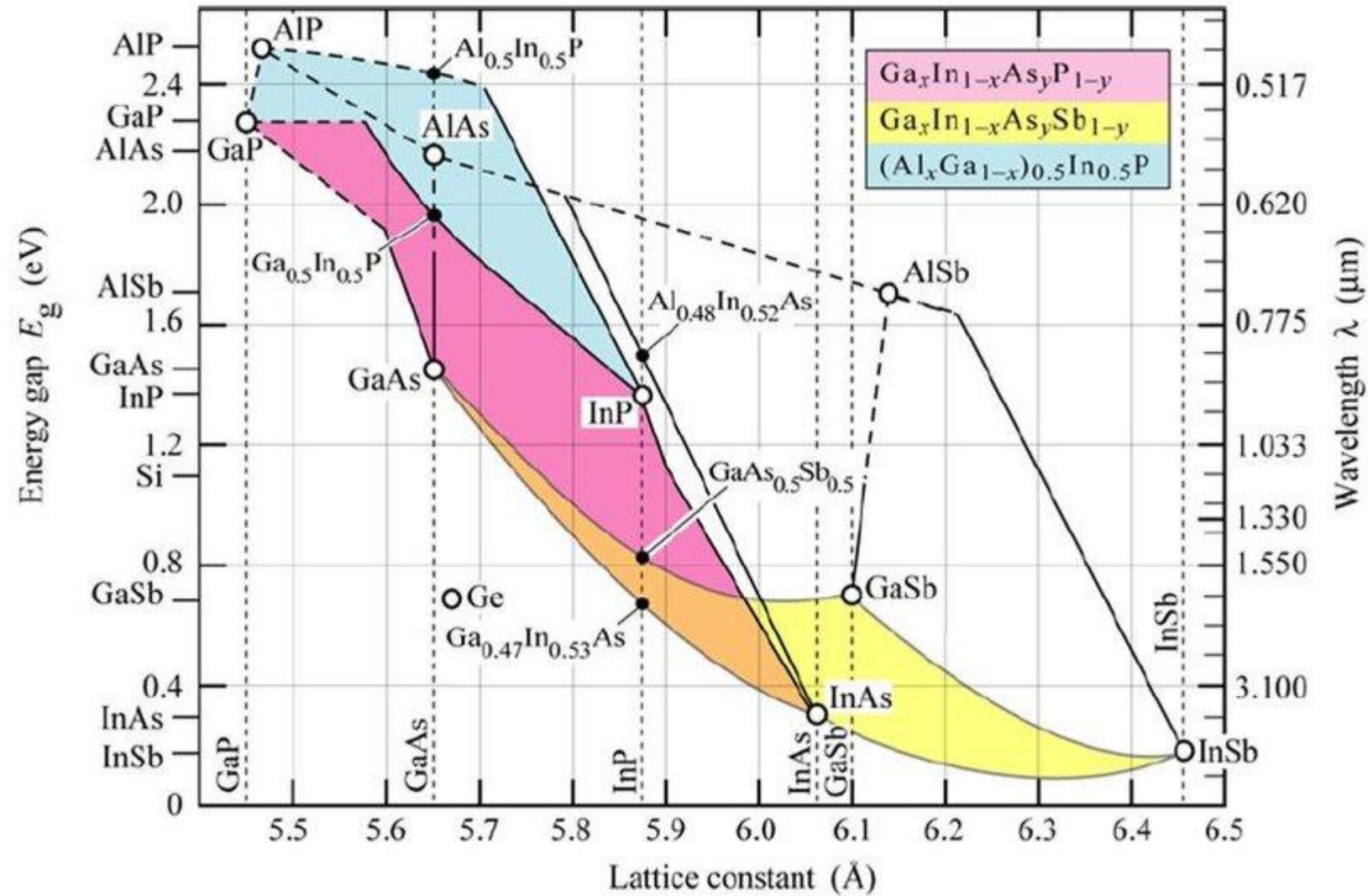
$$I_C \approx -\frac{qAD_n n_{p0}}{W_B} \left[\exp\left(\frac{V_{BE}}{V_T}\right) \right] \approx -\frac{qAD_n n_{i,B}^2}{W_B N_B} \left[\exp\left(\frac{V_{BE}}{V_T}\right) \right]$$
$$I_B \approx -\frac{qAD_p p_{n0}}{W_E} \left[\exp\left(\frac{V_{BE}}{V_T}\right) \right] \approx -\frac{qAD_p n_{i,E}^2}{W_E N_E} \left[\exp\left(\frac{V_{BE}}{V_T}\right) \right]$$
$$\beta = \frac{I_C}{I_B} = \frac{D_n n_{i,B}^2 W_E N_E}{D_p n_{i,E}^2 W_B N_B}$$

Strict requirements on base and emitter doping

1. Low base doping => increase in resistance, channel length modulation
2. Too high emitter doping => Bandgap narrowing => Reduction in carriers above barrier, Increase in C_j

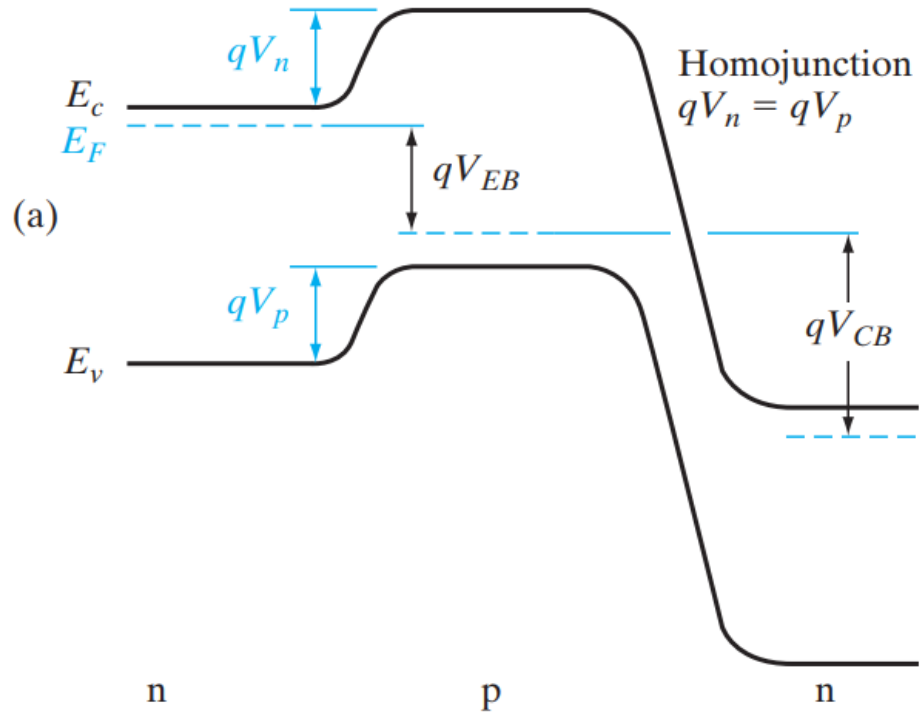
Can different bandgap materials be used for E, B and C?

Material Bandgap Dependence

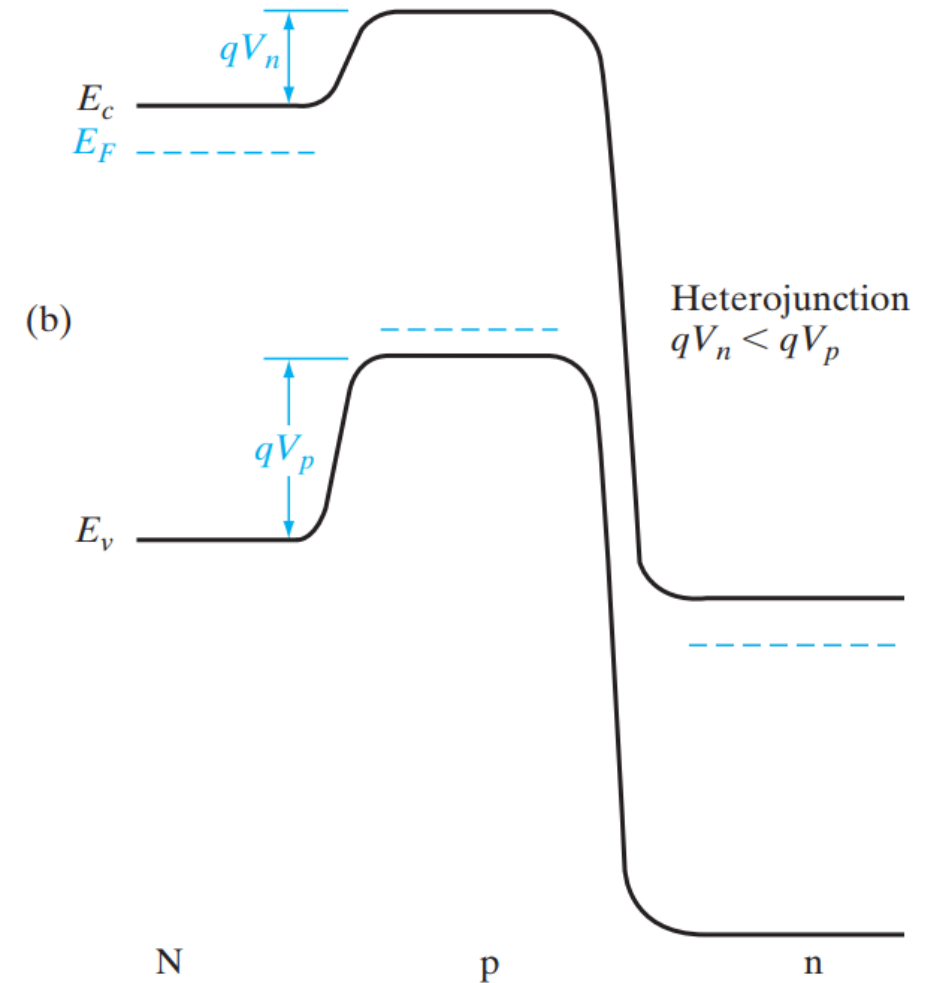


Homojunction vs Heterojunction Bipolar Transistor

Homojunction



Heterojunction



Heterojunction Bipolar Transistor

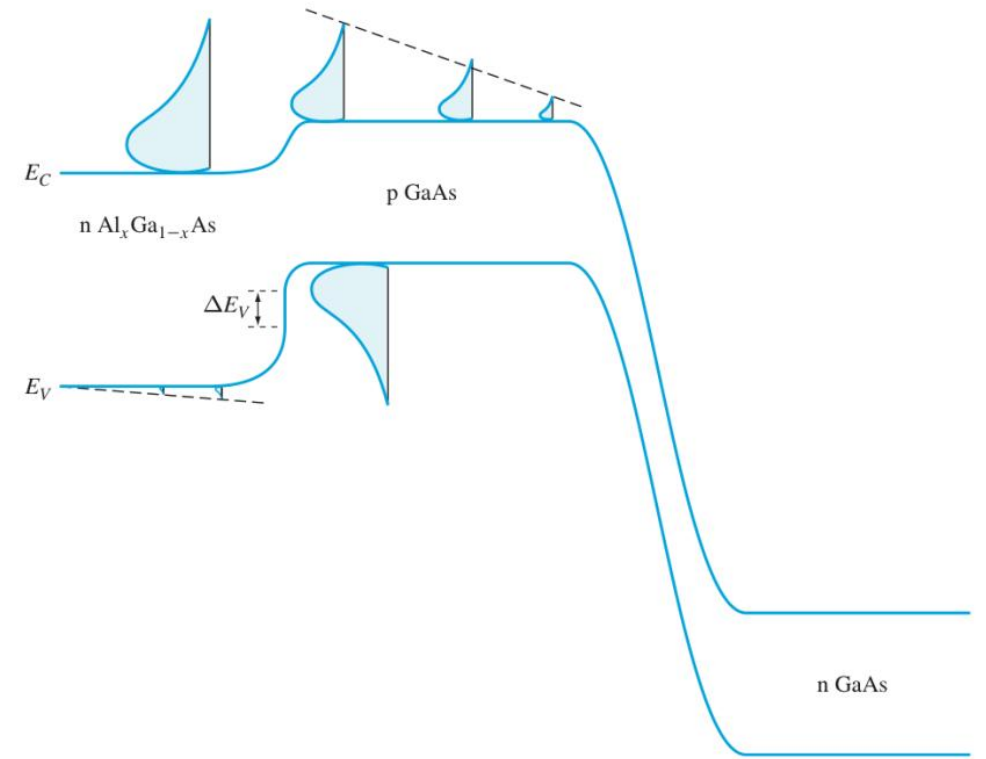
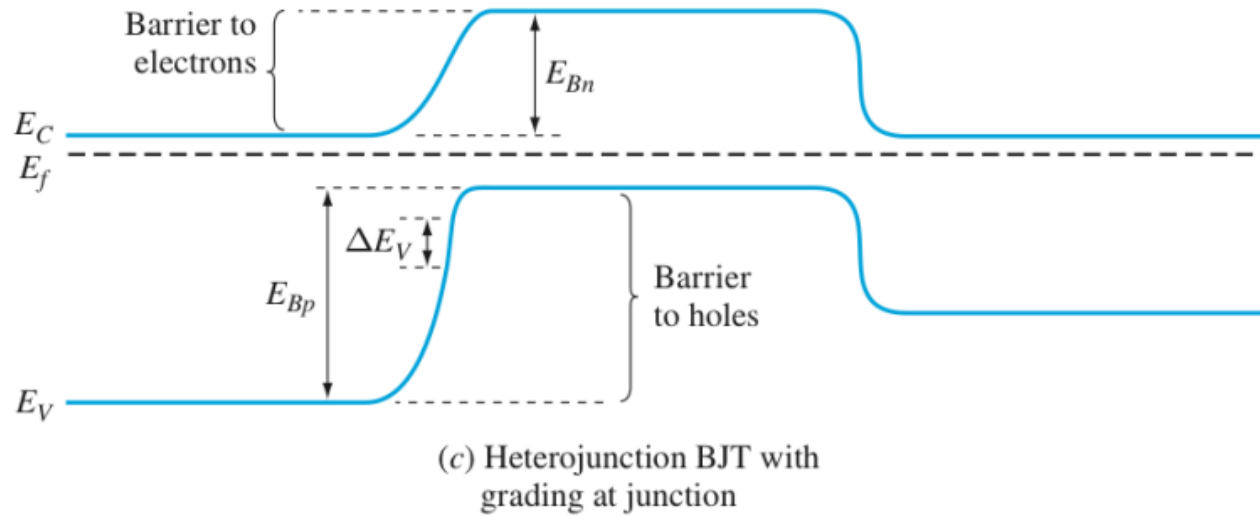
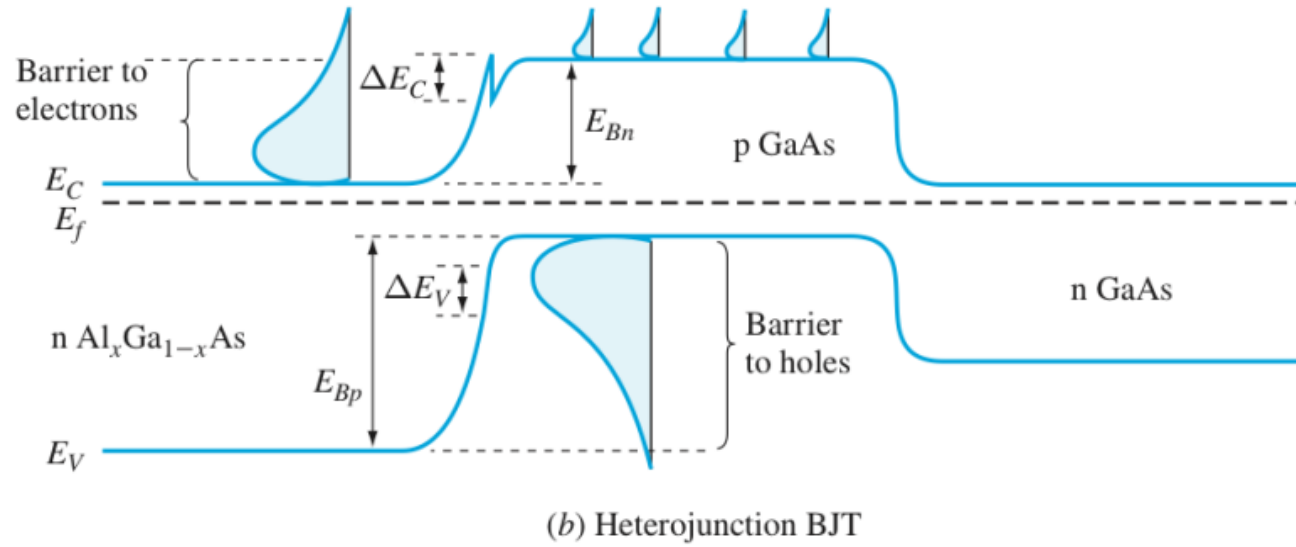


Figure 9.15 The HBT under forward active bias.